

Definizioni di AI literacy /3

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AI literacy si declina in ciò che gli studenti devono sapere e saper fare.

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Envisioning AI for K-12: What Should Every Child Know about AI?

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Abstract

The ubiquity of AI in society means the time is ripe to consider what educated 21st century digital citizens should know about this subject. In May 2018, the Association for the Advancement of Artificial Intelligence (AAAI) and the Computer Science Teachers Association (CSTA) formed a joint working group to develop national guidelines for teaching AI to K-12 students. Inspired by CSTA's national standards for K-12 computing education, the AI for K-12 guidelines will define what students in each grade band should know about artificial intelligence, machine learning, and robotics. The AI for K-12 working group is also creating an online resource directory where teachers can find AI-related videos, demos, software, and activity descriptions they can incorporate into their lesson plans. This blue sky talk invites the AI research community to reflect on the big ideas in AI that every K-12 student should know, and how we should communicate with the public about advances in AI and their future impact on society. It is a call to action for more AI researchers to become AI educators, creating resources that help teachers and students understand our work.

Introduction

Advances in AI have led to both beneficial and unintended negative effects, which have not escaped our research publications or mainstream media. For example: machine learning has greatly improved pattern recognition for medical diagnosis, but bias in a training set can lead to fairness issues in evaluating loan applications or problems with non-Caucasian facial recognition. These stories drive the imaginations and the fears of the general public. Despite the increased frequency of AI headlines in the media, there is a persistent lack of understanding of AI (West and

Allen 2018). We must consider the role we play as AI researchers and educators in helping people understand the science behind our research, its limits, and its potential societal impacts. On the graduate level, courses have been keeping pace with advances in the field. In undergraduate education, we've sought to provide fun and inspiring ways to engage students in various aspects of AI, such as robotics, modeling and simulation, game playing, and machine learning (Dodds, Hirsh, and Wagstaff 2018). In addition, we've sought to write textbooks and design tools, nifty and model AI assignments, and curricula that help students learn and apply the fundamentals of AI. More recently, companies have been making AI APIs such as IBM Watson, Google Web Speech, DialogFlow, and Azure Cognitive Services publicly available to educators and students.

Our hope for undergraduates is to prepare them for a bright future in the computing industry, or spur them on to graduate studies. But it is just as important now to think about what AI education should look like in K-12, not only to ensure a more informed populace that understands the technologies they interact with every day, but also to inspire the next generation of AI researchers and software developers. For many in this generation, AI will be an overlooked, magical force that powers their lives much as electricity, the internal combustion engine, and networking technology power ours.

Background

Over the past five years there has been a rapid expansion of CS education into formal K-12 curricula for all students, both nationally and internationally. In the United States this work has been undertaken by the National Science

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